

Pennywise, Power Foolish

A look at how to identify a good power-supply

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Being Agent001 brings along a lot of responsibilities that includes advising people while purchasing power supplies. Going away from the glamorous 3DTVs and exotic phones, my task this time is finding out more about power supplies. It's something most people seem to ignore. In fact, it's one of the last things people look for just after selecting what cabinet they want for their next build.

Instead of going on my usual trip to Lamington Road and finding the best deals on the power supplies, I decided to do some research to find out more about these less dazzling category of products. I wanted to answer a few commonly asked questions – how do I choose a good PSU? What do all these terms and standards, and do they really mean anything to us users?

One of the first questions that I'm asked is why should anyone even bother with specifically buying an expensive power supply, when the cheap ones work just fine? Often, the power supplies sold for low prices don't really match the requirements of the system. A 400-rupee 500W power supply is rarely 500W. In fact, it may be closer to 300W or 350W. The last thing a user wants is an under-performing power supply. If you are willing to take the risk, be prepared to witness BSODs, random reboots and sometimes shutdowns.

To counter this, people think that buying a PSU with the largest capacity is the way to go. The gamers amongst us have mid-range gaming systems with a mid-range graphics card such as the Radeon 5850 on a Core i5 system for



a 600W to 800W power supply. There's rarely the need for a 1000W or higher.

There are some who have many hard drives, and people are under the impression that hard drives consume a lot of power. It's not true. Most desktop drives consume around 4W on idle and 8W under load. That means you can easily run around eight hard drives on a decent 350W PSU running a basic system configuration.

Power supplies also have tons of features. There's also a lot of aggressive marketing that is done. One of these claims is the 80 PLUS rating. It started with a few brands and a few models but with so many levels, it's common to see not-so-great power supplies coming with these stamps. The 80 PLUS tag seen these days is a new standard started by a company that measures efficiency of a power supply at set power ratings. This is not to say that 80 PLUS is irrelevant. There are rumours of brands which are using the 80 PLUS labels on their products even without getting the certification.

There are a few other things vital to a good power supply – it has to be efficient, cool and reliable. The efficiency is important, but those who use a PC for a few hours in a day, it's not going to add up to a lot of savings. For more serious power users, it's important. Inefficient power supplies tend to be hotter and require better cooling, which in turn means faster and noisier fans. In the long run, these may not be very reliable either. PFC is another important part of a good power supply. It helps in improving the

example. In such cases, a stable 450W PSU should be more than sufficient. Again, we should emphasise that a cheap 400-rupee 400W PSU will not do. For those with the need to SLI or CrossFire, most two mid-range card configurations will be sufficiently powered by

efficiency of the power supply. Active PFCs are more effective than passive PFCs. Expect to pay more for these PSUs.

A lot of people like to gauge the quality of the power supply by the weight – the heavier the better. Connectors are features one should look at while purchasing a power supply. SLI users would want sufficient 6 and 8-pin connectors for the two or three graphics cards. Good quality cables is another thing to look out for. Well sheathed long cables with good quality connectors are important.

Now, for some recommendations based on past experience and from general opinion in terms of reliability. In today's markets, purchase 400W power supplies priced at above ₹1,500 at least. A good 450W PSU such as the Corsair VX450 would be available in the range of ₹3,500 and ₹4,000. Higher capacity models from Corsair's HX and TX series are also rated to be very good.



Similarly, Cooler Master's Real Power series is another range of power supplies. Seasonic is a brand that is known for its power supplies. In fact, they're the ones making power supplies for some of the top brands. The Corsair HX620 model is great for upper mid-range systems running a fast Core i7 processor along with a single a single fast card. For those looking for something little under the ₹6,000 price range, there is the HX520. They are modular power supplies are also decent for basic SLI/CrossFire setups. **d**

Calculate your requirements

Use Outervision's eXtreme PSU Calculator to get an idea on the capacity of the PSU required
<http://extreme.outervision.com>